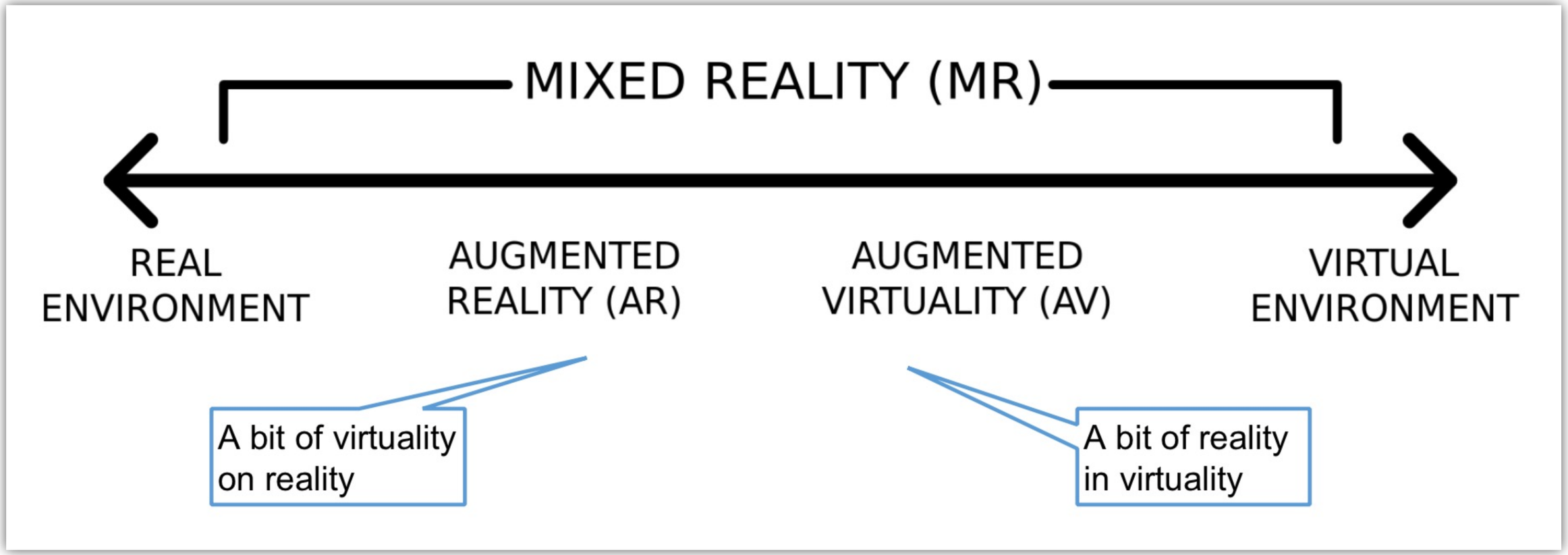
The Real and Virtual Worlds

Lecture 15

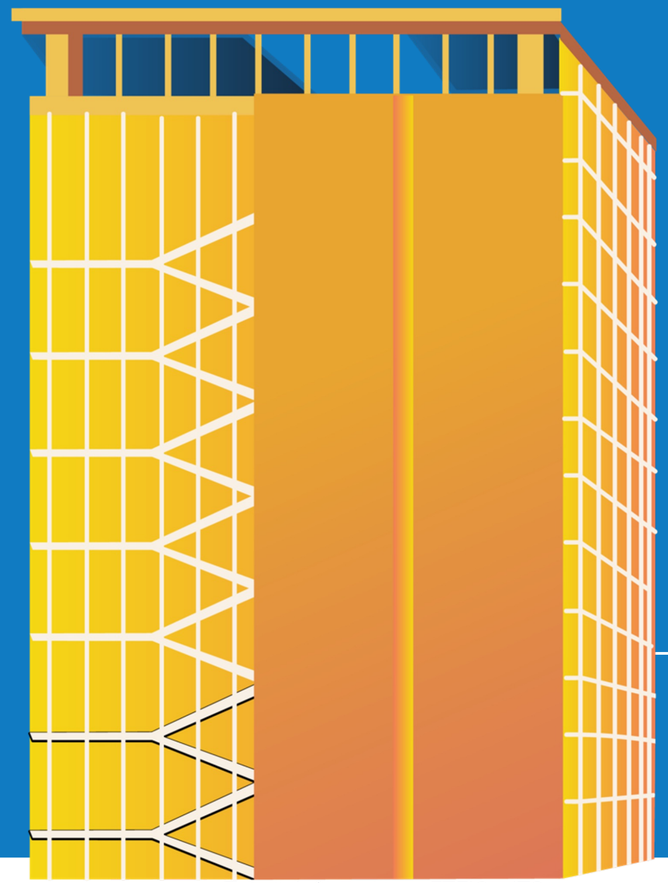
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Kontinum Realitas Virtualitas Milgram



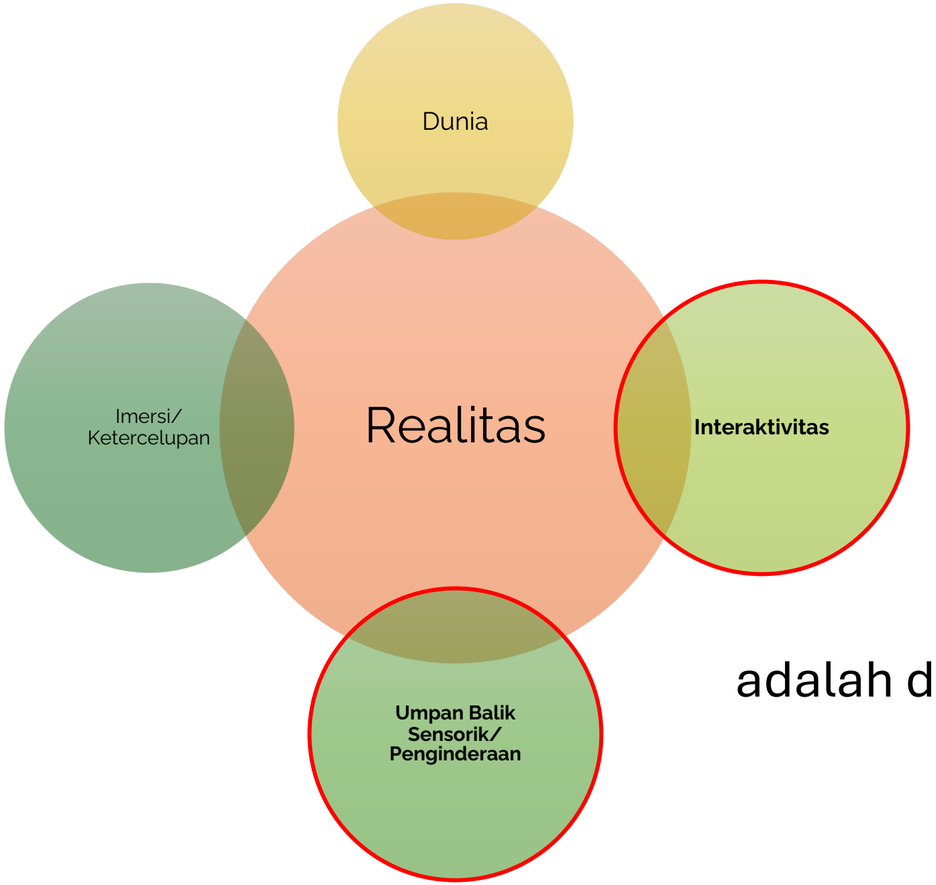
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Penginderaan dan Interaktifitas

Penginderaan dan Interaktifitas



Realitas merupakan sebuah lingkungan yang dialami melalui **rangsangan sensorik** (utamanya penglihatan dan pendengaran) yang mana aksi dari agen di dalamnya akan menentukan secara parsial apa yang akan terjadi selanjutnya di lingkungan tersebut.

adalah dua elemen utama di realitas.

Indra Manusia



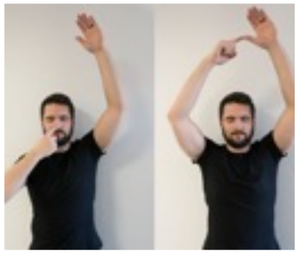
Penglihatan



Pendengaran



Sentuhan



Proprioception



Keseimbangan
atas Gerakan Fisik



Pembauan dan
Pengecapan

Menekan tombol

Menatapkan mata

Membuat gerakan

Berjalan

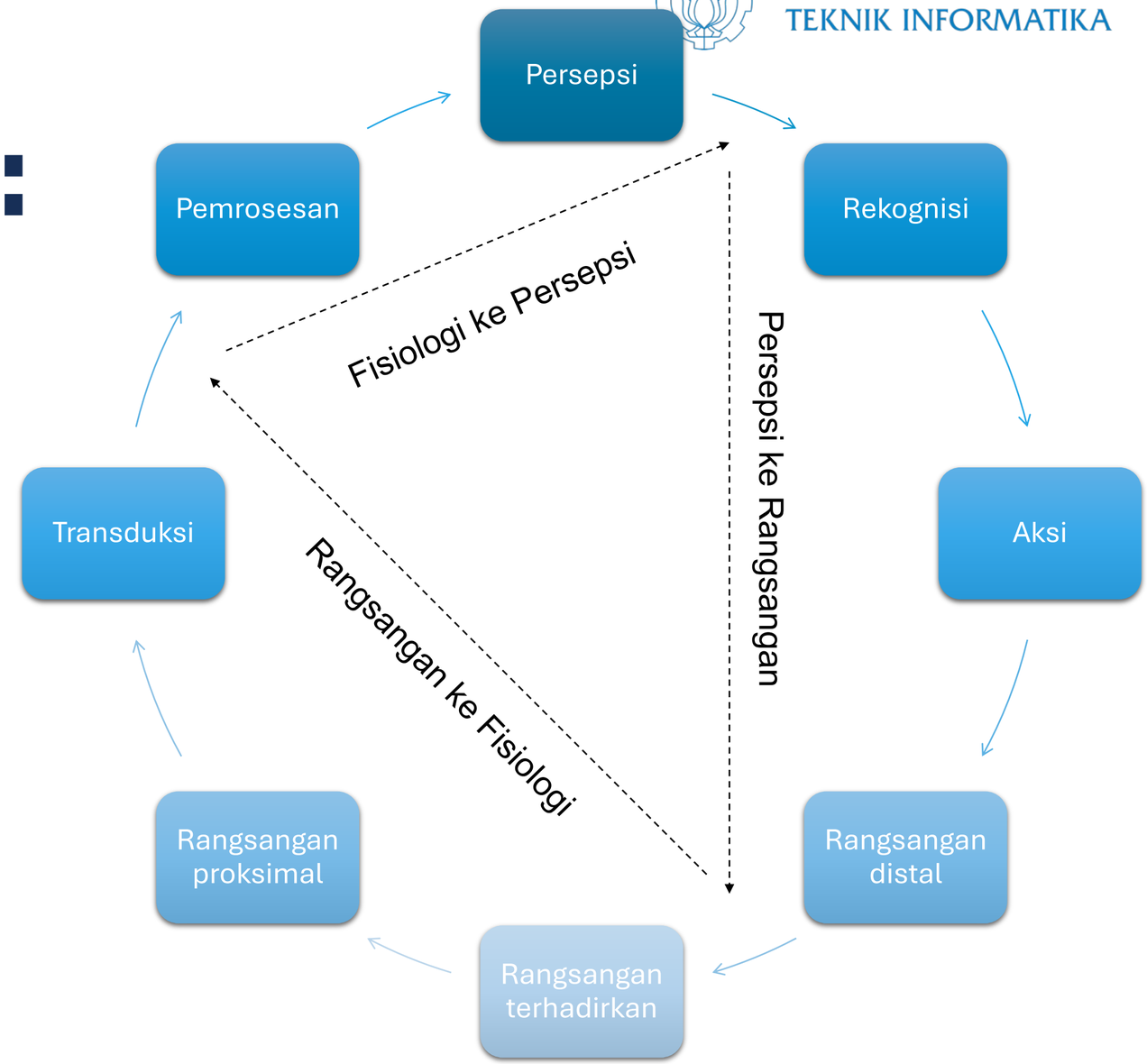
Berbicara

Proses Goldstein:

Melalui XR, kita membajak sistem **persepsi-aksi** manusia

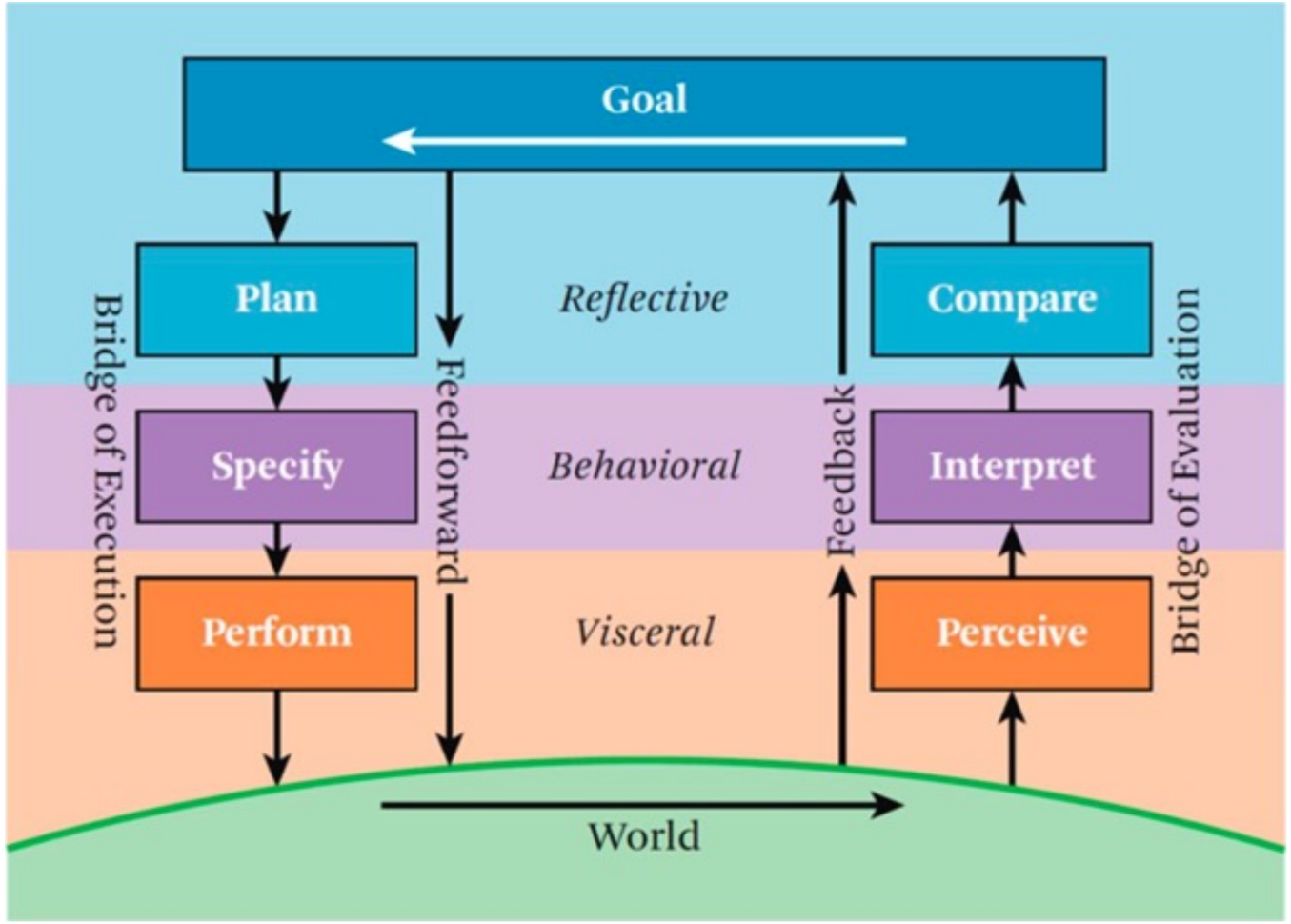
Aksi manusia = Input/Pelacakan XR

Rangsangan distal = Output XR



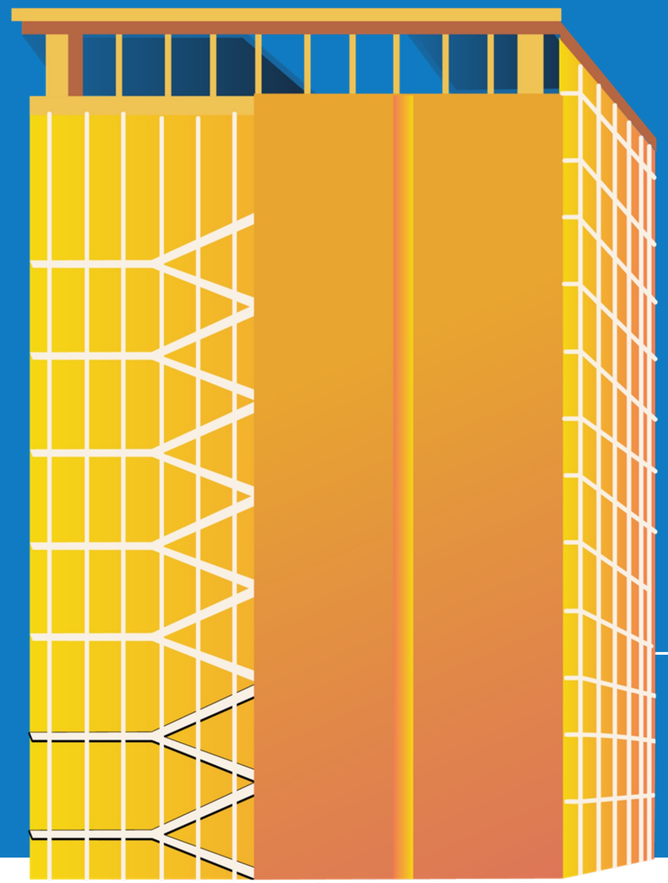
Perulangan Persepsi-Aksi

Siklus Interaksi



diadaptasi dari
Siklus Interaksi Norman

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Antarmuka dan Dunia Virtual

Antarmuka (Interface) dan Dunia Virtual (Environment)







https://www.uxmatters.com/mt/archives/2020/08/new-realities-mobile-vr-design.php
https://developer.qualcomm.com/blog/xr-user-interfaces-and-perception-technologies

Elemen dalam Rancangan XR

- [1] Comfort and Safety
- [2] Interaction (affordance, signifiers, feedback)
- [3] Environment and spatial components
- [4] Sensory Input
- [5] Engagement (storytelling, gamification)
- [6] Constraints
- [7] Inclusion, diversity, and accessibility

Emerging VR Conventions

Curved Screens	Diegetic UI First	Long Press Button	Comfort Zone	The Throwaway – delete-gesture	Stereoscopic Consistency
VR Smart Watch Menu	Hand Interaction	Normal Maps	Locomotion, Snap Rotation, Motion Sickness Option	Constant Velocity	VR Onboarding
Text Readability	VR Button Convention	Selection Ray Function	HUD Motion Lag	VR Menus	Legacy Interaction – Gaze Activation

Thank You

Tim RMK GIGA

